EDA Report : Answering Business Questions

PART 1: Observations & Anomalies

Leads :

```
Total Number of Unique Applicants: 15114
Total Number of Unique Callers: 12
Total Number of Unique Countries : 140
Total Number of Unique Campaign IDs: 23
Total Number of Unique Campaign Names: 23
Total Number of Enrolled Applicants: 104
Total Number of Unique Enrolled Students: 104
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Reachable Vs Non-Reachable & Duplicates Vs Non _ Duplicates

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Total Applicants - 33069
Total Unique Applicants - 15114
Total Reachable Leads (Connected): 8879
Total Un-reachable Leads (Not Connected):6635
Reachability Rate: 58.75%
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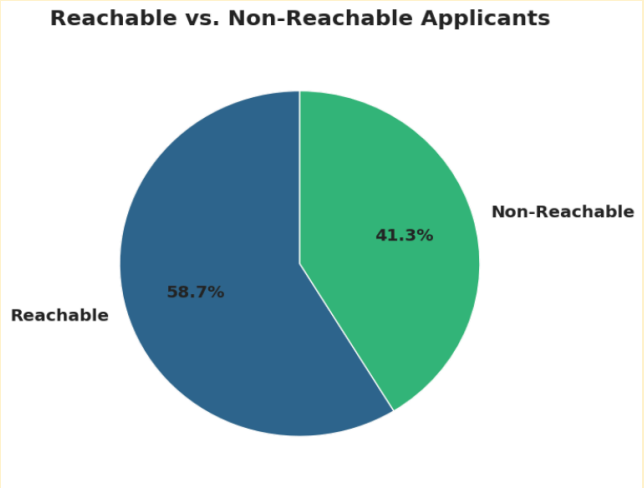

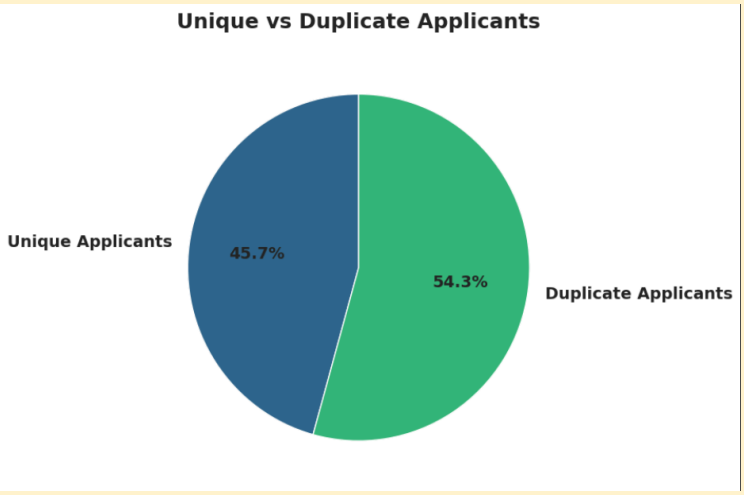

- The Reachability Gap: Around 41.3% of the total lead database (6,653 out of 15,114 applicants) never transitioned into a successful contact, representing a significant waste of marketing spend.
- Significant gap between Unique and Duplicate Applicants suggests that there were multiple calls made to the unique applicants with an average of 2 calls per unique applicant, the team is heavily invested in a 'nurturing' sales cycle rather than a high-volume, single-touch approach to bring in more students for the University of Illinois.

Distributions :

```
Call_Status
Not Connected    69.35%
Connected        30.65%

Escalation_Required
No     97.74%
Yes     2.26%
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Admission_Category
Post Admission    78.61%
Pre Admission     21.39%
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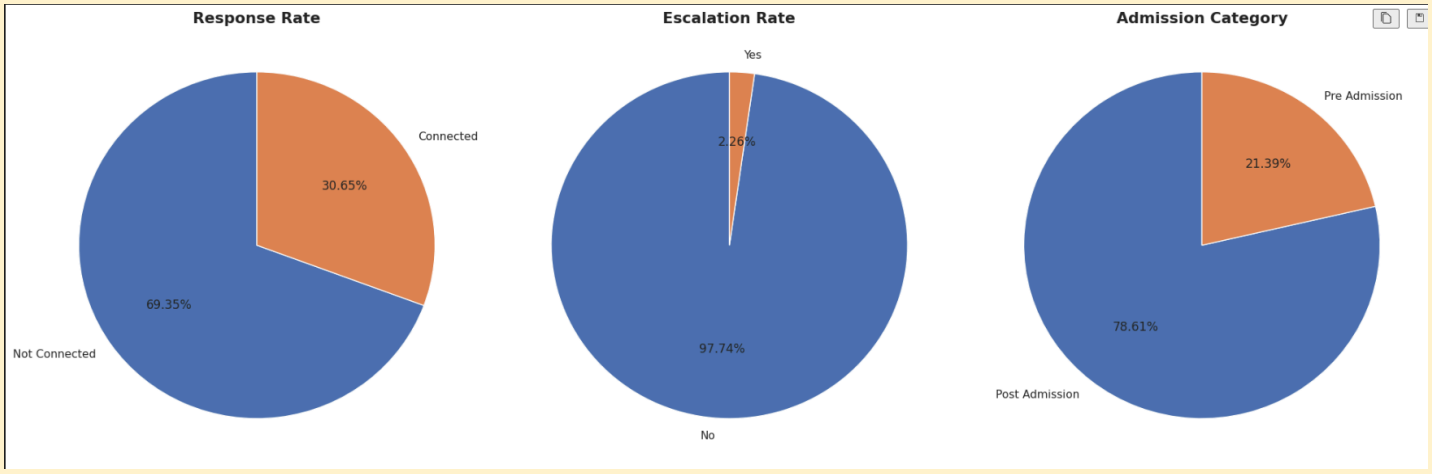
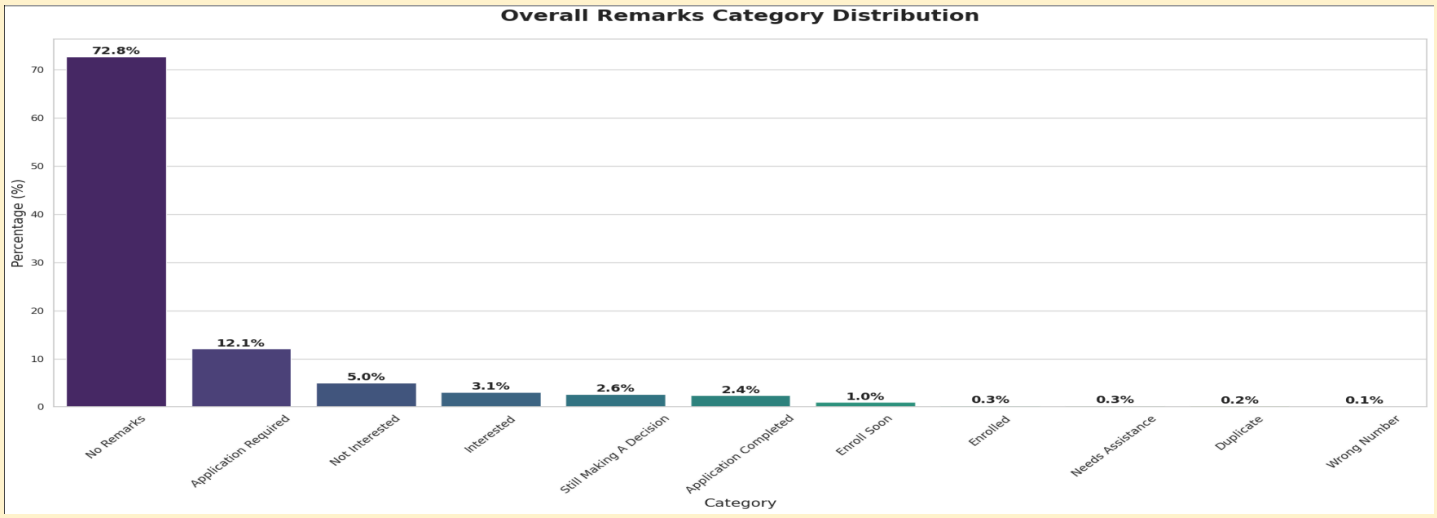


Nearly 70% of outreach fails to connect, indicating significant resource wastage on unreachable leads or follow-ups because of unsettled queries. However, 98% of calls require no escalation, showing high agent autonomy. Surprisingly, 79% of activity targets Post-Admission students, suggesting the team prioritizes administrative onboarding over new "Pre-Admission" sales conversion.

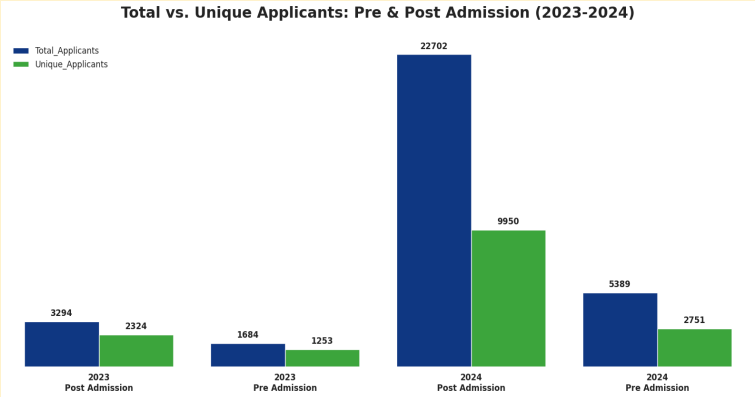
Remarks Category Distribution Rate :

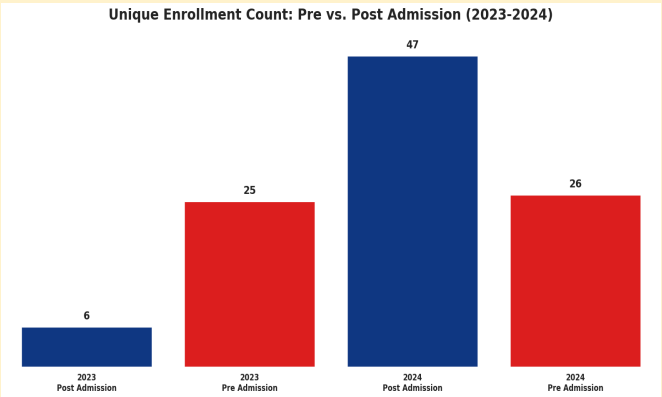


- Despite 79% of outreach efforts being dedicated to Post-Admission support, the final enrollment rate stands at a marginal 0.3%. This indicates that the team's high 'Support Volume' is not translating into 'Final Success,' suggesting a need to simplify the final enrollment steps or shift focus back to high-intent Pre-Admission leads.
- Reporting Complexity: Initial attempts to visualize all remark categories resulted in extreme label overcrowding, indicating a very high variety of specific outcomes that require high-level categorization (like Remarks_Category) for meaningful "big picture" analysis.

PART 2: Outreach Operational Performance Analysis

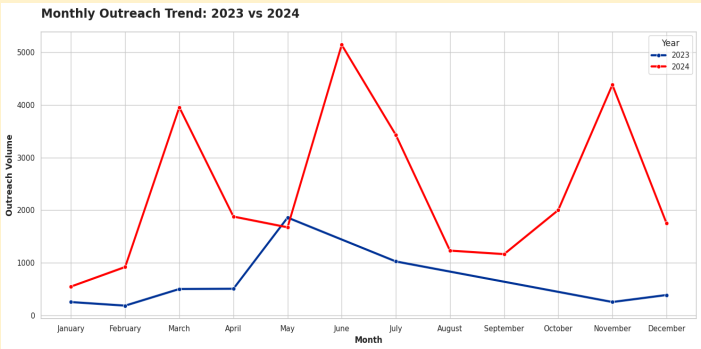
Pre & Post Admission - 2023 & 2024 :

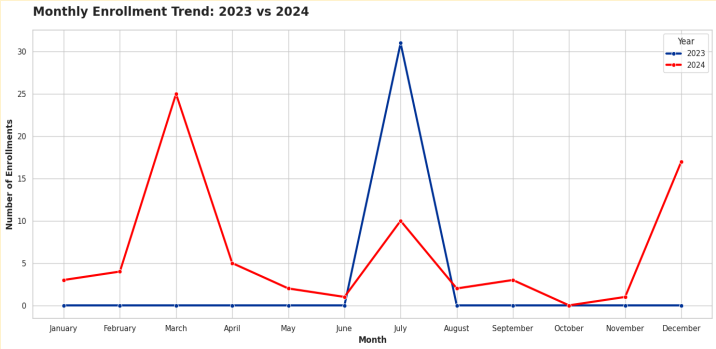




- Pre-Admission Efficiency: Top-of-funnel engagement with new applicants demonstrates superior conversion, yielding significant enrollment volume with lower relative effort.
- Post-Admission Diminishing Returns: Massive outreach volume in post-admission fails to drive proportional enrollment.
- Conversion Barriers: Stagnation in post-admission results is attributed to external factors like visa delays, competitive offers, and unresolved technical queries.
- Resource Optimization: Shifting focus toward new lead acquisition is more cost-effective than exhaustive & un-necessary follow-ups on low-intent applicants.

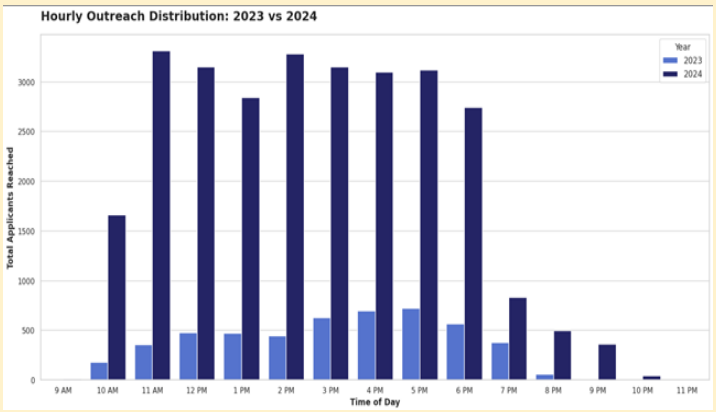
Monthly Outreach & Enrollment Trend :

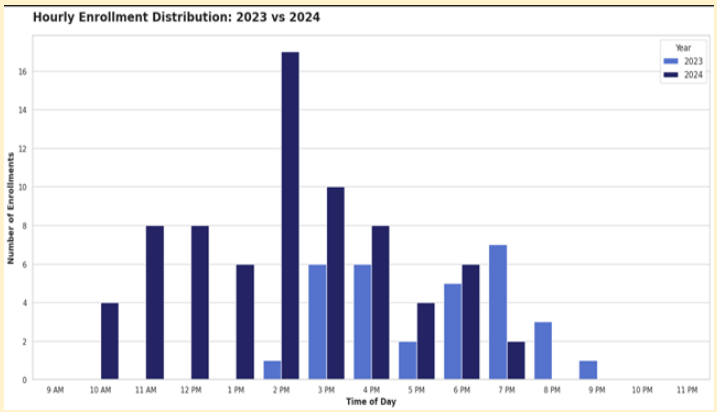




- Outreach Volume vs. Conversion: 2024 outreach surged significantly, peaking in March (~4000), June (~5,200) and November (~4,400), yet these spikes did not yield immediate enrollment.
- Post-Admission Impact: Despite the massive volume of 22,702 post-admission outreach attempts in 2024, enrollment was primarily driven by high-intent "Unique Applicants" and support calls.
- Strategic Enrollment Shifts: 2023 relied on a singular July peak (~31)because of pre-admission new applicant targets, whereas 2024 shows diversified conversion cycles in March and December.
- Priority Realignment: While pre-admission outreach remains efficient for lead acquisition, the 2024 data suggests that post-admission query resolution is critical to locking in final enrollments.
- Operational Conclusion: To maximize ROI, resources should pivot from broad volume-based outreach to finding newer applicants and providing high-touch support for existing applicants to resolve conversion barriers like visa or application issues.

Hourly Outreach & Enrollment Trend :



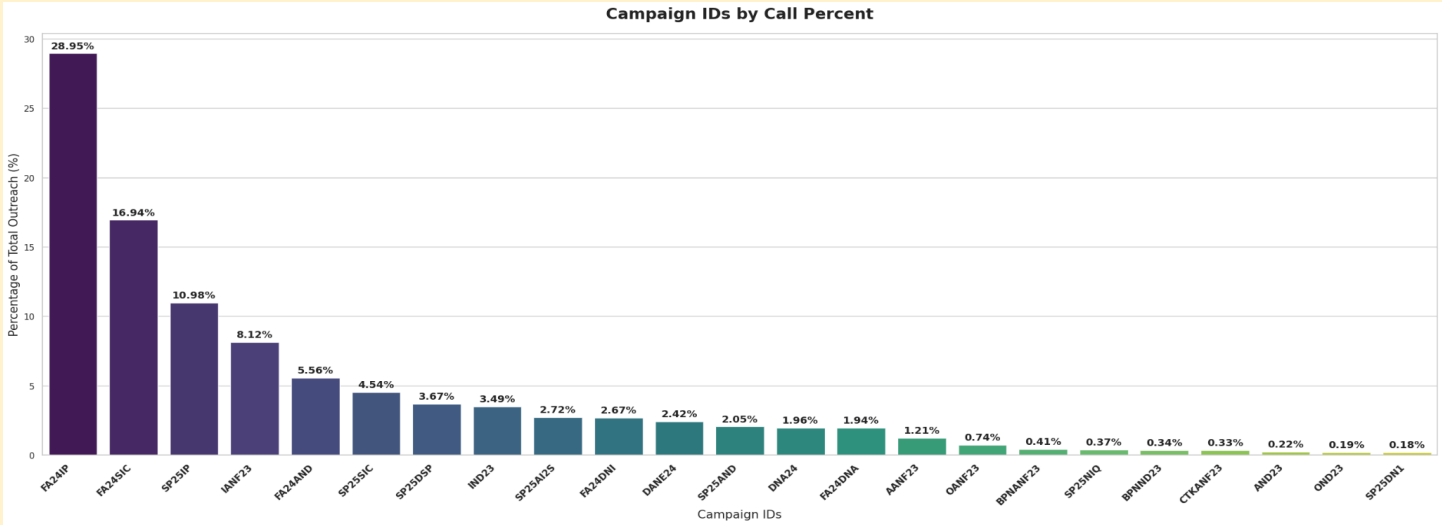


- 2024: Outreach and enrollment metrics show strong alignment, with a clear performance peak at 2 PM within a broader 10 AM–6 PM active window.
- 2023: Engagement trends were decoupled; while outreach was steady throughout the afternoon, the highest enrollment density occurred during the late-day 3 PM–8 PM period.
- Increasing call volume definitely is insufficient as it depends more on the type of calls made (newer targets or query calls) and type of applicant outreached, like we see in the data for pre and post admission.

PART 3: Campaign & Category Effectiveness Analysis

Campaign ID - Outreach Rate :

The outreach is heavily concentrated in three campaigns—FA24IP, FA24SIC, and SP25IP—which together account for nearly 57% of all calls. Conversely, high-efficiency campaigns like AND23 receive less than 1% of total outreach, suggesting a significant misalignment between resource allocation and conversion performance.



Campaign ID - Return on Effect [ROE] by Enrollment vs Performance :

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--- Campaign Effectiveness Audit ---
            Total_Calls  Unique_Students  Enrolled_Students  \
Campaign_ID
AND23                73               59                  6
FA24AND            1838              810                 40
SP25AI2S            899              396                 18
IND23              1153              926                 23
OND23                62               42                  1
SP25AND             678              251                  4
DANE24              800              565                  7
BPNND23             114               83                  1
DNA24               649              484                  2
FA24DNA             641              340                  1

            Unique_Enrolled_Students  Intensity_Ratio  Lead_Conversion_Rate
Campaign_ID
AND23                              6             1.24                 10.17
FA24AND                           40             2.27                  4.94
SP25AI2S                          18             2.27                  4.55
IND23                             23             1.25                  2.48
OND23                              1             1.48                  2.38
SP25AND                            4             2.70                  1.59
DANE24                             7             1.42                  1.24
BPNND23                            1             1.37                  1.20
DNA24                              2             1.34                  0.41
FA24DNA                            1             1.89                  0.29

            Call_Efficiency_Rate
Campaign_ID
AND23                       8.22
FA24AND                     2.18
SP25AI2S                    2.00
IND23                       1.99
OND23                       1.61
SP25AND                     0.59
DANE24                      0.88
BPNND23                     0.88
DNA24                       0.31
FA24DNA                     0.16
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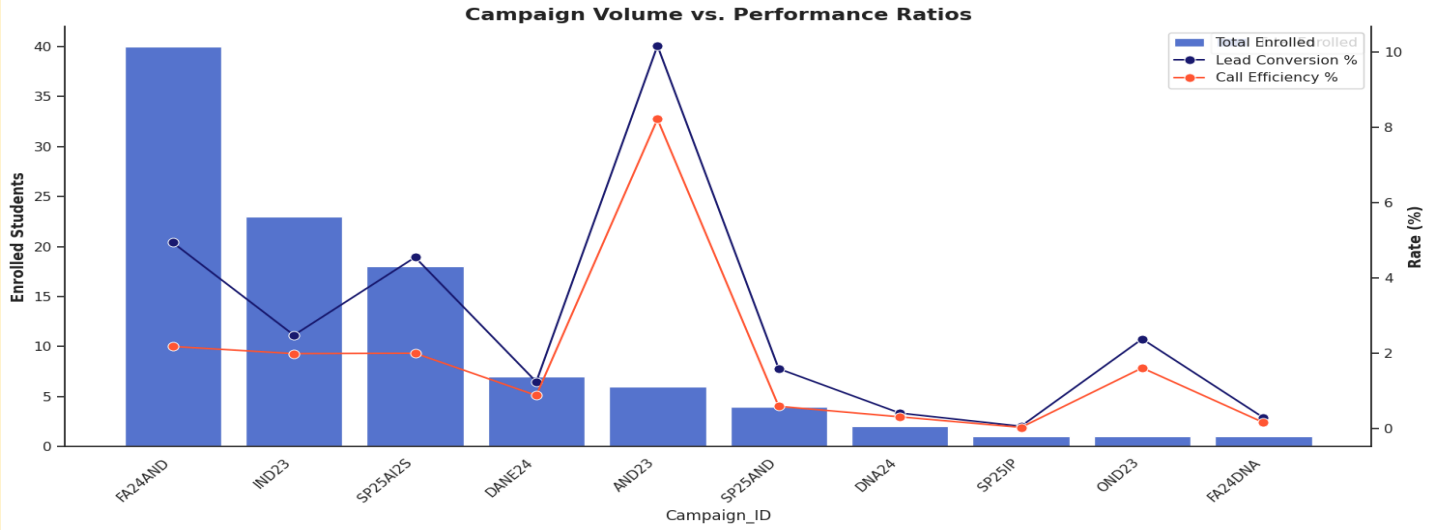


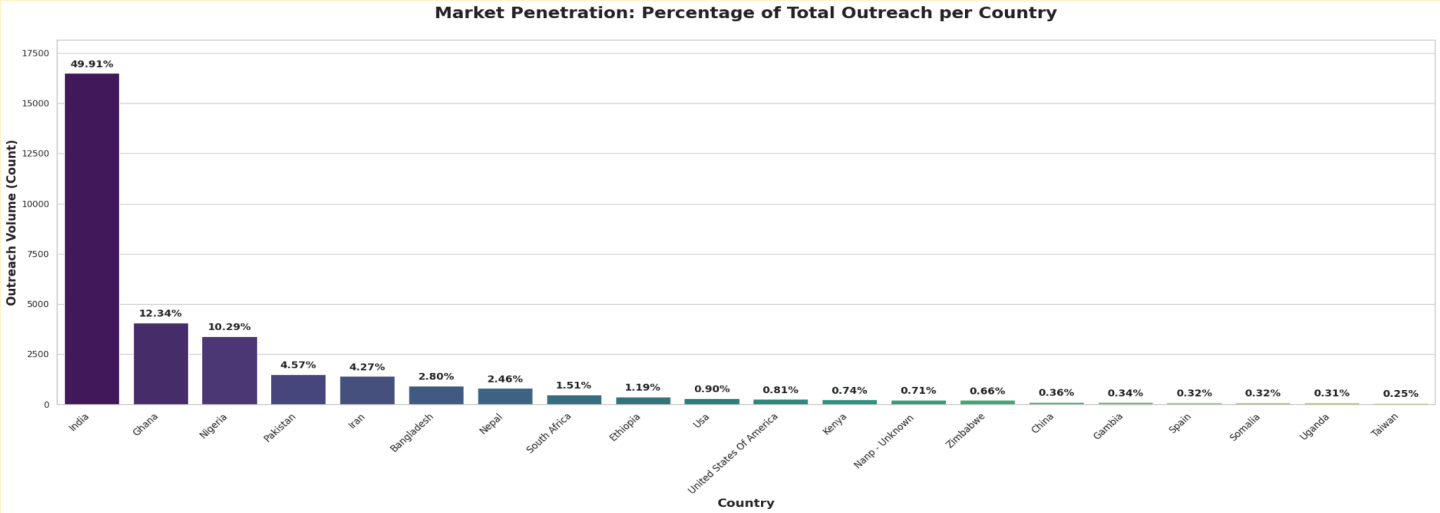
The data highlights a clear trade-off between high-volume reach and conversion efficiency across campaigns:

- While FA24IP, FA24SIC, and SP25IP—which together account for nearly 57% of Outreach, final enrollment coming from them stands to 0 %.
- Efficiency Leader: AND23 is the top performer in quality, boasting the highest Lead Conversion Rate (10.17%) and Call Efficiency Rate (8.22%). Despite having only 73 total calls, it yielded 6 enrollments, demonstrating a highly optimized targeting strategy.
- Volume Driver: FA24AND generated the most scale with 40 enrolled students. However, this required a massive effort of 1,838 calls, resulting in a much lower Call Efficiency of 2.18%.
- Underperformers: Campaigns like DNA24 and FA24DNA show significant friction. Despite high call volumes (over 640 each), they produced minimal results, with Lead Conversion Rates plummeting below 0.40%.

- Consistency: SP25AI2S and IND23 maintain a healthy middle ground, securing 18–23 enrollments with steady conversion rates between 2.48% and 4.55%.
- Conclusion: The audit & chart suggests that the tactics used in AND23 should be analyzed and scaled, while the low-efficiency, high-outreach methods of the DNA series require immediate optimization.

PART 4: Strategic Market & Geographic Performance Analysis

Outreach per Country :



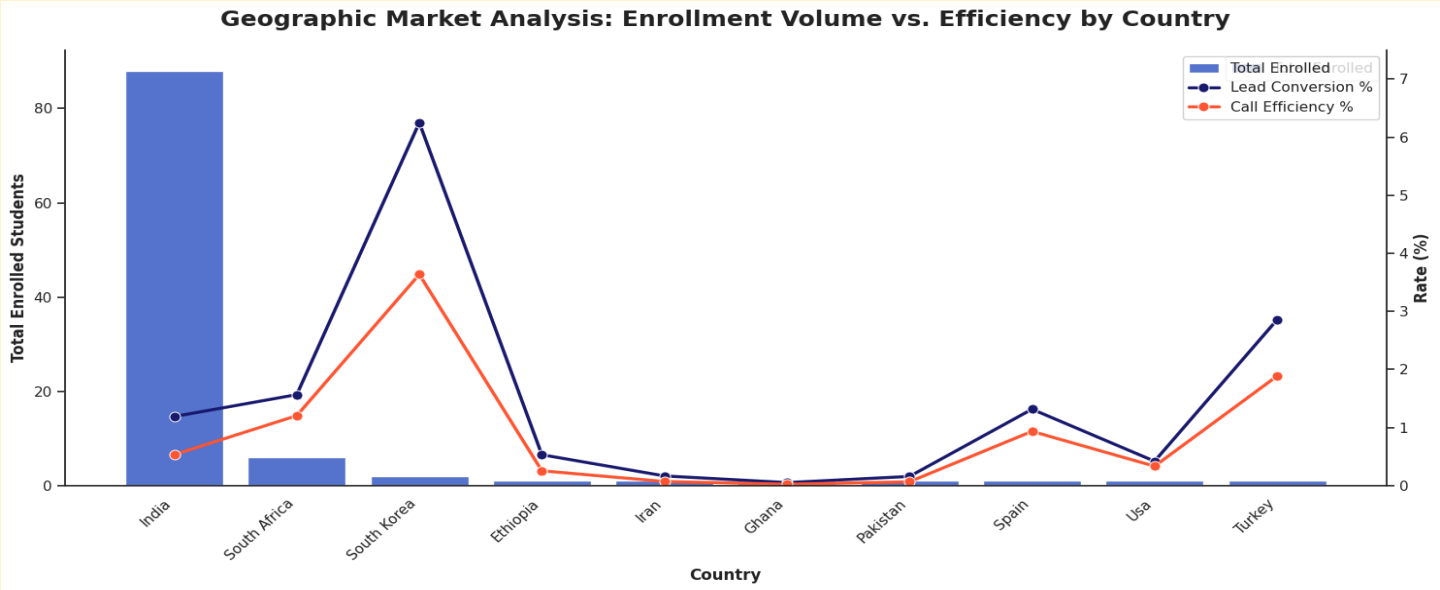
- The campaign exhibits a heavy concentration risk, with Developing Nations driving the vast majority of outreach volume. India (49.91%), Ghana(12.34%), and Nigeria(10.29%) collectively account for over 70% of total engagement, signaling high demand and strong brand resonance within these emerging markets.
- Conversely, penetration in Developed Nations (USA, China, Spain, Taiwan) remains marginal, each contributing less than 1%. This disparity suggests the university's value proposition is currently optimized for students in developing economies seeking upward mobility, while highlighting a significant growth opportunity to diversify recruitment strategies in higher-income, developed regions to balance the portfolio.

Geographic Nation Analysis - Efficient Markets :

Country	Total_Calls	Unique_Students	Enrolled_Students
South Korea	55	32	2
Turkey	53	35	1
South Africa	498	383	6
Spain	107	76	1
India	16506	7395	88
United States Of America	269	175	1
Ethiopia	392	188	1
Usa	299	238	1
Iran	1411	599	1
Pakistan	1510	632	1

Country	Unique_Enrolled_Students	Intensity_Ratio \
South Korea	2	1.72
Turkey	1	1.51
South Africa	6	1.30
Spain	1	1.41
India	88	2.23
United States Of America	1	1.54
Ethiopia	1	2.09
Usa	1	1.26
Iran	1	2.36
Pakistan	1	2.39

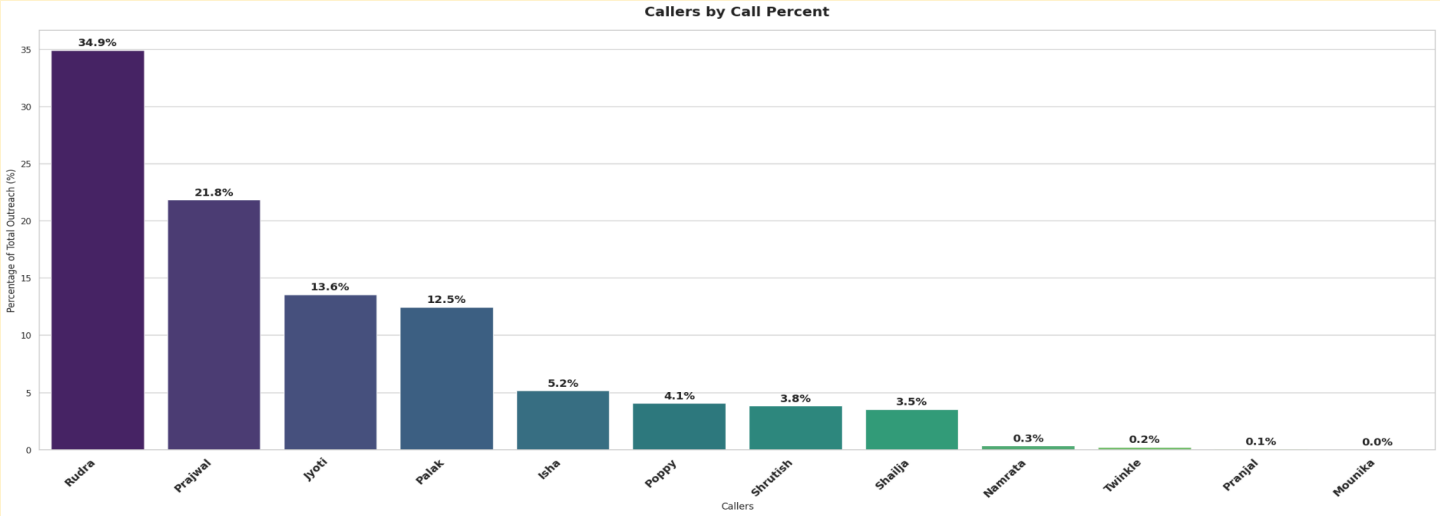
Country	Lead_Conversion_Rate	Call_Efficiency_Rate
South Korea	6.250000	3.636364
Turkey	2.857143	1.886792
South Africa	1.566580	1.204819
Spain	1.315789	0.934579
India	1.189993	0.533139
United States Of America	0.571429	0.371747
Ethiopia	0.531915	0.255102
Usa	0.420168	0.334448
Iran	0.166945	0.070872
Pakistan	0.158228	0.066225

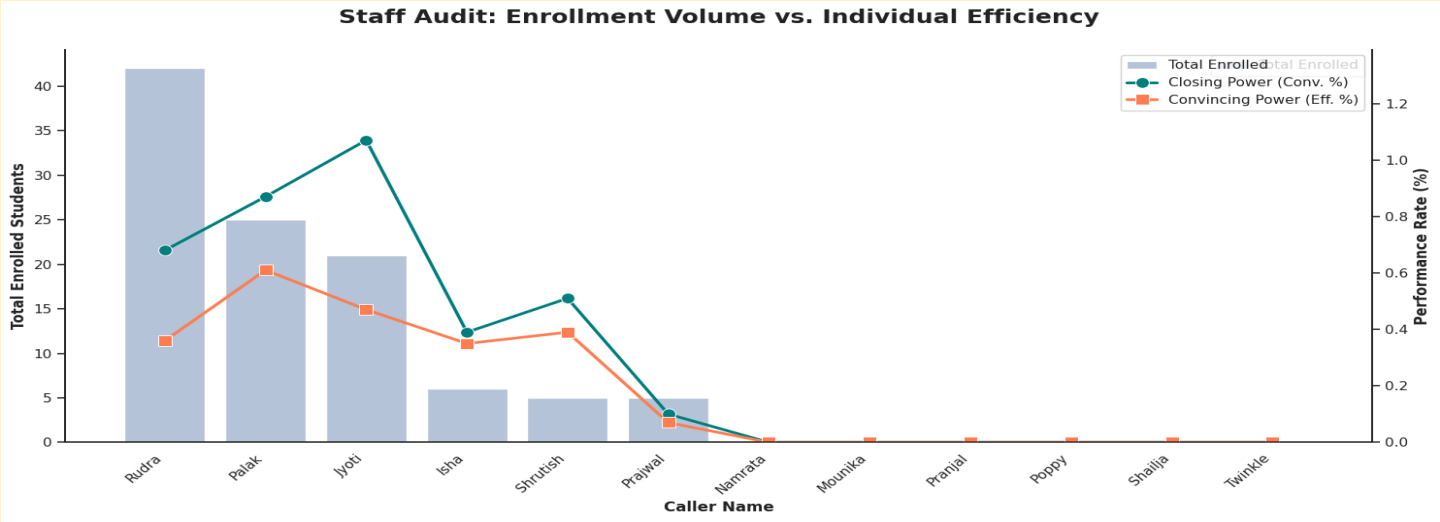


- The data reveals a stark inverse correlation between volume and efficiency. India dominates enrollment volume with 88 students, yet operates at a low Call Efficiency (0.53%) and Lead Conversion (1.19%). Conversely, South Korea emerges as the highest-performing market, achieving a peak Lead Conversion Rate of 6.25% and Call Efficiency of 3.64% despite lower enrollment volume.
- Developing markets like Iran and Pakistan represent significant operational leakage, maintaining high Intensity Ratios (>2.3) but yielding 1-2 conversion.
- Resource reallocation [increase in newer applicants + outreach volume] toward high-efficiency regions like South Korea and Turkey is recommended to optimize ROES (Return on Expense Spend).

PART 5: Call Performance & Caller Outreach Velocity Analysis

Callers Performance & Resource Optimization Insight :



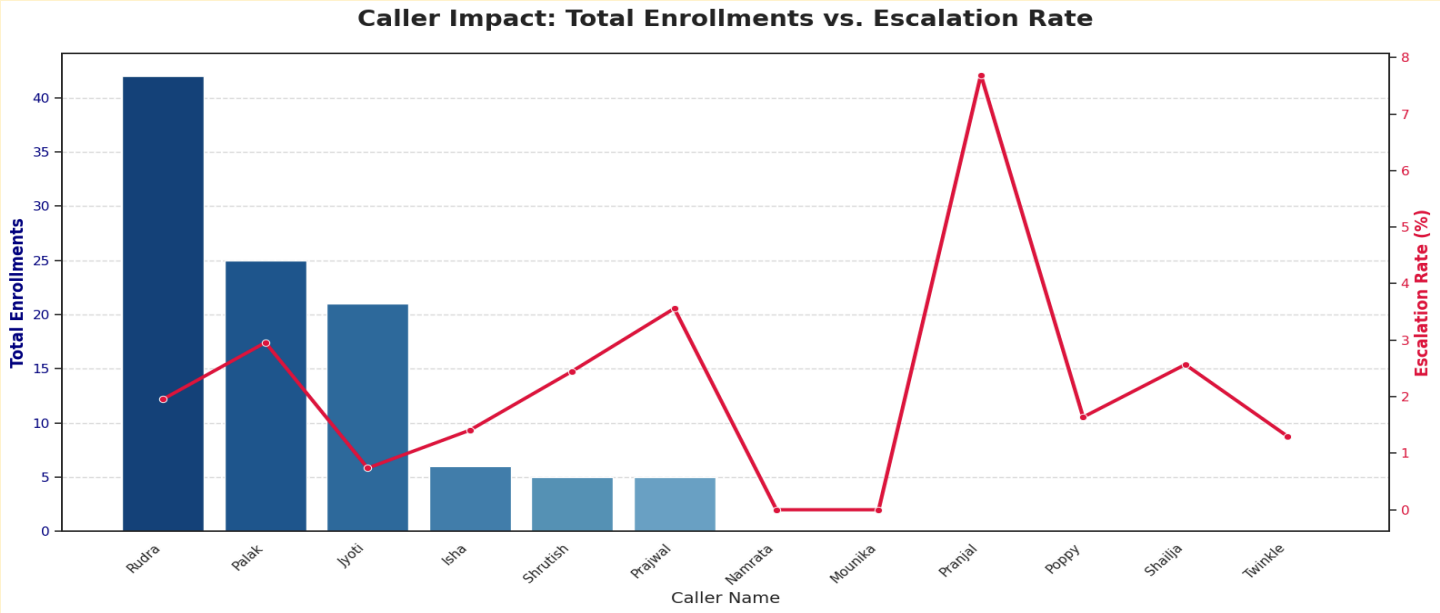


The audit reveals a significant misalignment between outreach volume and conversion efficiency. Rudra dominates total activity with a 34.9% outreach share and 42 enrollments, yet maintains a sub-optimal Closing Power (~0.7%). In contrast, Jyoti exhibits superior Outreach Velocity, achieving a peak Lead Conversion (Closing Power) of ~1.1% despite a significantly lower outreach footprint of 13.6%.

Key Performance Gaps:

- Operational Leakage: Prajwal represents high risk, consuming 21.8% of call volume with negligible conversion, indicating a critical need for re-skilling in lead nurturing.
- Resource Attrition: Staff members such as Mounika, Pranjal, Namrata, and Twinkle contribute near-zero ROI, suggesting their current roles are economically unviable.
- Training Opportunity: Poppy and Shailja show baseline activity but lack the "Convincing Power" to secure enrollments; they require intensive coaching to improve their Call-to-Enrollment ROI.
- Strategic Recommendation: Reallocating high-intent leads to high-efficiency closers like Jyoti and Palak to maximize ROAS and overall enrollment yield.

Escalation vs. Enrollment

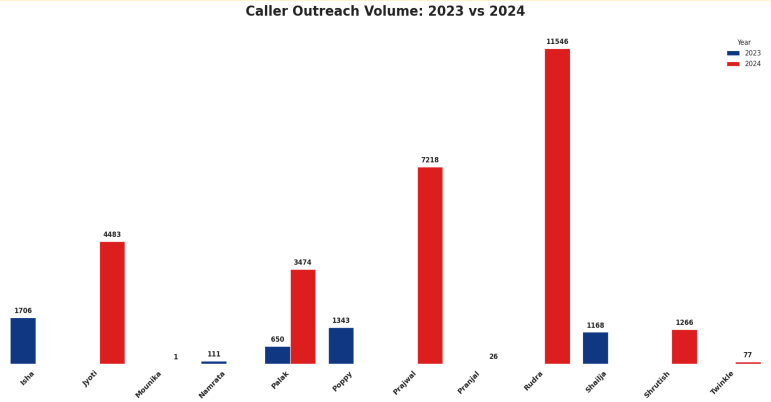


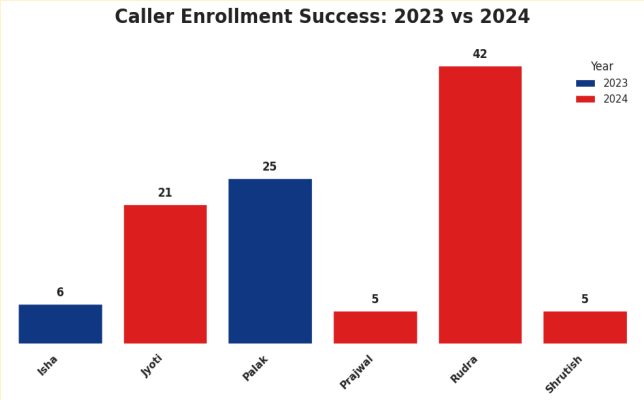
The Escalation Rate serves as a critical proxy for Lead Friction and Closing Difficulty. The data identifies Jyoti as the most efficient performer, maintaining a high enrollment count (21 students) with the lowest Escalation Rate (< 1 %), signaling high autonomy and effective communication.

Key Performance Corridors:

- High-Volume Friction: Rudra drives the most enrollments (42) but at a moderate escalation cost (2%), suggesting that scaling volume naturally increases support complexity.
- Efficiency Red Flag: Prajwal shows a high Escalation Rate (3.5%) alongside low enrollments (5), indicating that her outreach efforts are frequently stalling and requiring intervention.
- Outlier Inefficiency: Pranjal exhibits a peak Escalation Rate of nearly 8% with zero enrollments, representing a significant Resource Drain that likely negatively impacts the university's brand reputation.

Callers Performance Analysis: 2023 vs. 2024 Outreach & Enrollment :

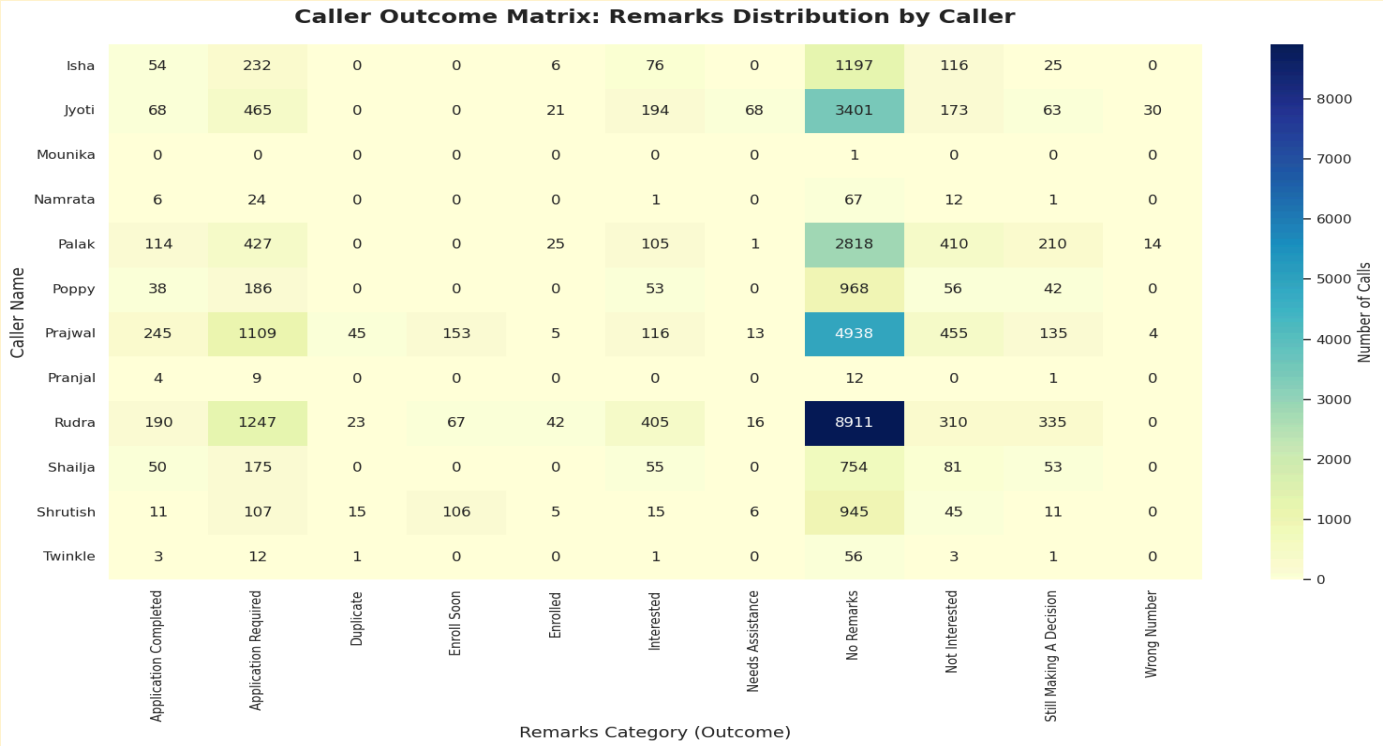




- 2023 operations were primarily sustained by Isha, Palak, Poppy, and Shailja, with Palak emerging as the anchor, delivering a peak of 25 enrollments from a relatively low outreach volume.

- Conversely, 2024 performance is driven by massive scaling efforts from Rudra, Prajwal, and Jyoti. Rudra dominates the current period with a high-volume strategy (11,546 calls), yielding a team-leading 42 enrollments. However, Jyoti demonstrated higher conversion efficiency in 2024, securing 21 enrollments with nearly a third of Rudra's call volume. Shrutish and Twinkle appear as 2024 additions with high Operational Leakage, contributing to volume but failing to match the enrollment success of 2023's legacy performers.

Caller Activity vs. Outcome Distribution :



The Caller Outcome Matrix provides a critical diagnostic view of sales activity and data integrity, revealing that "No Remarks" is the primary outcome across all top callers.

Key Operational Insights:

- Documentation Gap: A massive volume of calls - led by Rudra (8,911), Prajwal (4,938), and Jyoti (3,401), lack any qualitative remarks. This represents a significant loss of business intelligence and suggests a need for stricter CRM logging protocols.
- Conversion Quality: Rudra and Prajwal show high activity in "Application Required" and "Application Completed," yet their enrollment counts remain disproportionately low compared to their total call volume. Applicants might have changed their decisions at the end or need more follow ups.
- Lead Stagnation: Prajwal has the highest count of "Application Required" (1,109), indicating a large pool of leads that are interested but failing to cross the finish line into "Enrolled" status, probably need more future follow ups.
- Niche Success: Palak and Jyoti maintain more balanced distributions, with significant numbers in "Interested" and "Application Completed" relative to their smaller outreach footprint, confirming their higher Call Efficiency.